MZUZU UNIVERSITY

Department of Mathematics

MATH 1101 - Precalculus

Supplementary Examination

Monday, June 12 2017 **Time:** 8 : 00 – 11 : 00 **(3 hours)**

1. Check that the paper has **12 questions**.
2. Attempt all questions
3. Write answers in the space provided
4. Show all your working clearly
5. Use proper notation
6. Simplify your answers as far as possible
7. **No Calculators**

1. (a) Use properties of exponents in computing the following quantity. (2)

$$\left(\frac{144 \cdot 125}{2^3 \cdot 3^2}\right)^{-1}$$

(b) Rationalise the denominator and simplify if possible. (3)

$$\frac{3}{\sqrt[4]{3}}$$

2. Present the following numbers on a number line $\sqrt{2}, e, \pi - 5, \ln e$. (2)

3. Find the perimeter of the triangle formed by joining the mid-points of of the sides of ΔABC with vertices $A(1,1), B(9,3)$ and $C(3,5)$. (5)

4. Determine whether the following pair of equations form lines that are parallel, perpendicular or neither; (2)

$$3x + 2y = 6; 5x + 6y = 30$$

5. Let $f(x) = ax^2 + bx + c$. Find (4)

$$\frac{f(x+h) - f(x)}{h}$$

6. Determine the domain of $h(x) = \frac{x}{\sqrt{6x^2+x-1}}$ present it using interval notation. (6)

7. Use translations and reflections to sketch the graph of $y = \sqrt{-x+3}+2$, specifying intercepts. (5)

8. (a) Evaluate $\log_2 8\sqrt{2}$. (3)

(b) Solve the equation for x: $3(2^{2x}) - 11(2^x) - 4 = 0$ (4)

(c) Solve the equation for x: $\log_{\sqrt{2}}(1 - 4x) = 4 + \log_{\sqrt{2}}(1 + 4x)$. (4)

9. Specify the domain of $\log_{10}(x^2 - 6x - 6)$ and hence solve the inequality $\log_{10}(x^2 - 6x - 6) > 0$, presenting your solution in interval notation. (7)

10. Express each term in terms of sines and cosines and show that (3)

$$1 - \frac{\cot^2 \theta}{\csc^2 \theta} = \sin^2 \theta$$

11. (a) Convert $\frac{11\pi}{6}$ radians to degrees. (2)

(b) Convert $450°$ to radians. (2)

12. (a) If $\sec t = -\frac{5}{3}$ and $\pi < t < \frac{3\pi}{2}$, compute $\cos t$ and $\tan t$. (4)

(b) Evaluate $\sec\left(-\frac{5\pi}{3}\right)$ (4)